

2.1 Introduction of Vector-Valued Functions

Definition: A vector-valued function or vector function, is simply a function whose domain is a subset of the real numbers and whose range is a set of vectors. If the vector function $\mathbf{r}(t)$ maps real numbers to n -dimensional vectors, we write: $\mathbf{r}: \mathbb{R} \rightarrow \mathbb{R}^n$.

Note: In this class we are most interested in vector functions of the form $\mathbf{r}: \mathbb{R} \rightarrow \mathbb{R}^3$.

Note: Very strictly speaking, it is a slight abuse to write $\mathbf{r}: \mathbb{R} \rightarrow \mathbb{R}^n$ when the domain of $\mathbf{r}$ is not the full set of real numbers. In particular if D is the domain of $\mathbf{r}$ and it is a subset of the real numbers but not equal to the real numbers, we should write $\mathbf{r}: D \rightarrow \mathbb{R}^n$. We will ignore this subtlety in this class and be aware that $\mathbf{r}: \mathbb{R} \rightarrow \mathbb{R}^n$ only means that the domain of $\mathbf{r}$ is a subset of real numbers.

Definition: If a vector function is of the form $\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$, then $f(t)$, $g(t)$, and $h(t)$ are called the component functions of $\mathbf{r}$.

Note: In this case we have $\mathbf{r}: \mathbb{R} \rightarrow \mathbb{R}^3$ because the real number t gets mapped to a vector $\langle f(t), g(t), h(t) \rangle$.

Example 1: Find the domain of the vector function: $\mathbf{r}(t) = \langle \sqrt{4-t^2}, \cos(t), \sqrt{t^2-1} \rangle$.

Note: It is important to see that as we graph a vector-valued function in this class we are simply graphing a parametric function in 3-space. Therefore, t will act as a parameter and each vector-valued function will have an associated orientation.

Example 2: We have seen that the parametric equations for the line that passes through the points (2,-3,4) and (4,2,1) are: $x = 2 + 2t$ $y = -3 + 5t$ $z = 4 - 3t$. How could one represent this line as a vector function?

Example 3: Sketch a graph of each of the following functions.

a) $\mathbf{r}(t) = (t^2 - 2t + 1)\mathbf{i} + (t - 1)\mathbf{j}$ for $0 \leq t \leq 3$.

b) $\mathbf{r}(t) = 4\cos(t)\mathbf{i} + 2\sin(t)\mathbf{j}$ for $0 \leq t \leq 2\pi$.

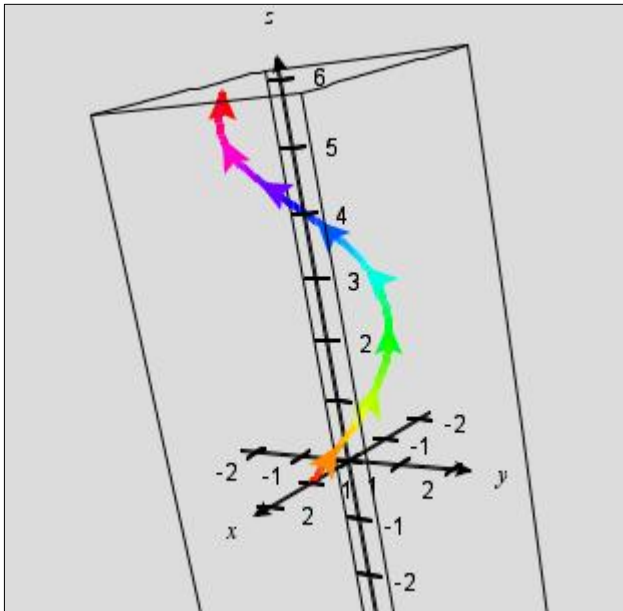
c) $\mathbf{r}(t) = 2\cos(t)\mathbf{i} + 2t\mathbf{j} + \sin(t)\mathbf{k}$ for $t \geq 0$.

Example 4: Use a computer to graph the vector function:

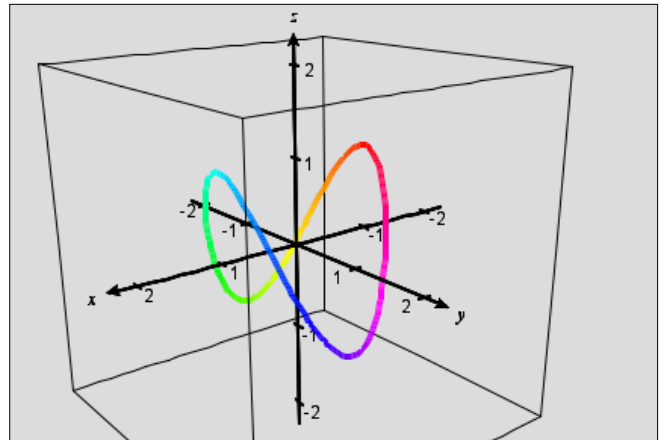
$$\mathbf{r}(t) = \langle (4 + \sin(20t))\cos(t), (4 + \sin(20t))\sin(t), \cos(20t) \rangle$$

Example 5: Match each picture below with its appropriate vector function (you are not allowed to use a computer).

Graph A:



Graph B:



Possible functions:

1. $\mathbf{r}(t) = \langle \cos(t), \sin(t), t \rangle$
2. $\mathbf{r}(t) = \langle \cos(t), \sin(t), \cos(2t) \rangle$
3. $\mathbf{r}(t) = \langle \cos(t), t, \sin(t) \rangle$
4. $\mathbf{r}(t) = \langle \cos(t), \cos(t), \cos(t) \rangle$